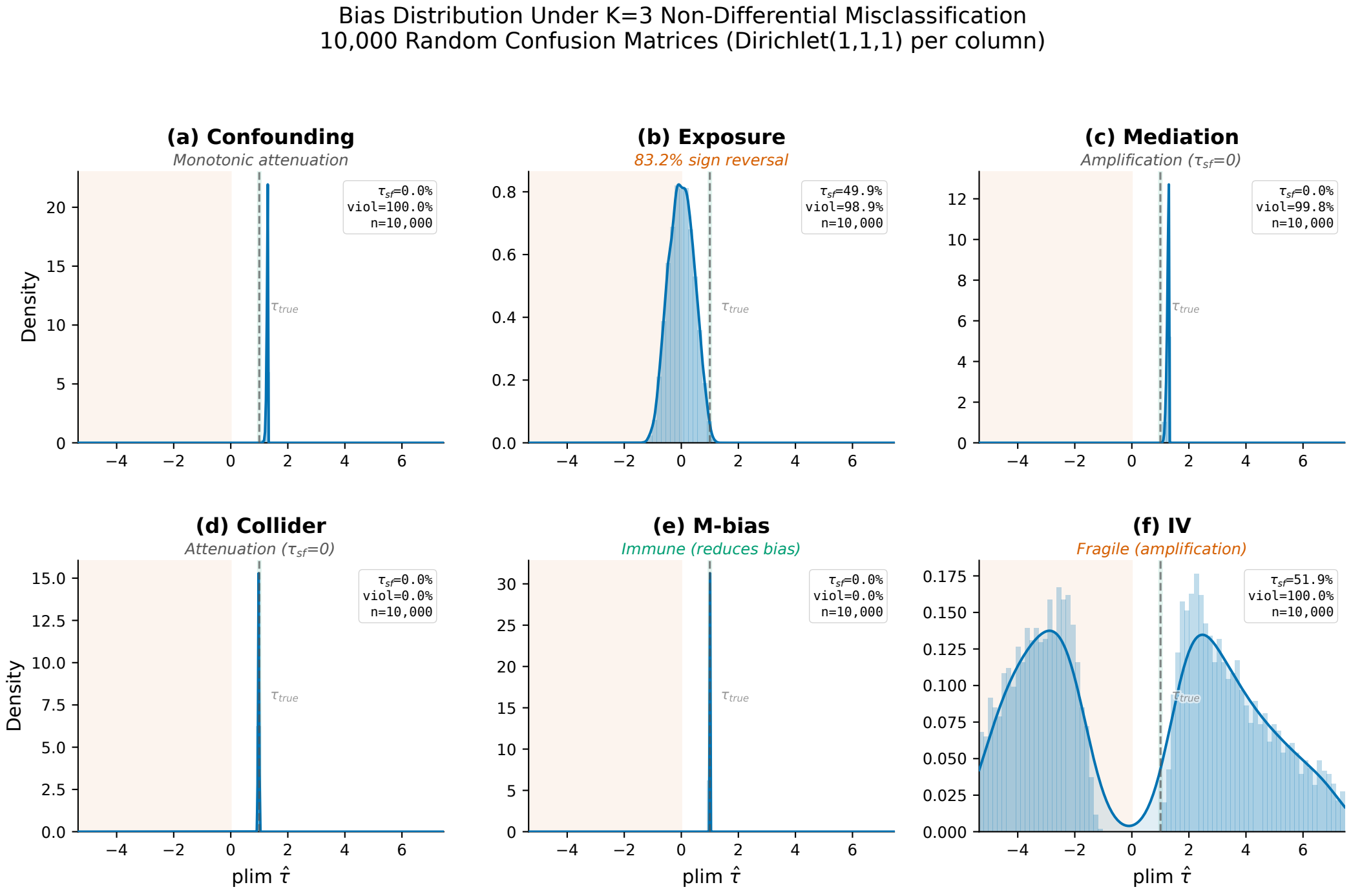

Bias Distribution Under K=3 Non-Differential Misclassification
10,000 Random Confusion Matrices (Dirichlet(1,1,1) per column)
(a) Confounding
Monotonic attenuation
τsf=0.0%
viol=100.0%
n=10,000
τtrue
(b) Exposure
83.2% sign reversal
τsf=49.9%
viol=98.9%
n=10,000
τtrue
(c) Mediation
Amplification (τsf=0)
τsf=0.0%
viol=99.8%
n=10,000
τtrue
(d) Collider
Attenuation (τsf=0)
τsf=0.0%
viol=0.0%
n=10,000
τtrue
(e) M-bias
Immune (reduces bias)
τsf=0.0%
viol=0.0%
n=10,000
τtrue
(f) IV
Fragile (amplification)
τsf=51.9%
viol=100.0%
n=10,000
τtrue
Density
plim τ̂